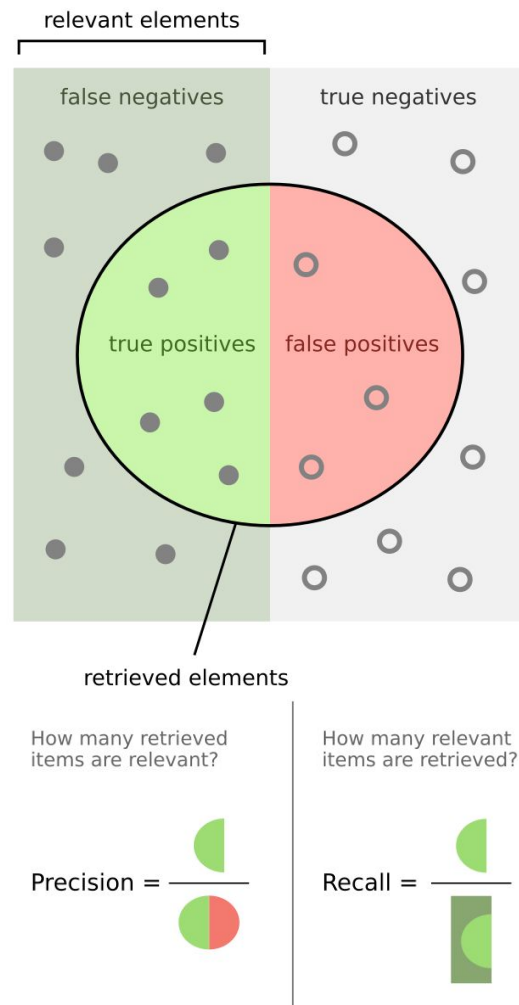


CTC TAB Update

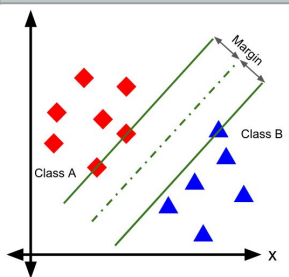
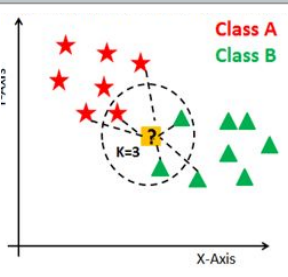
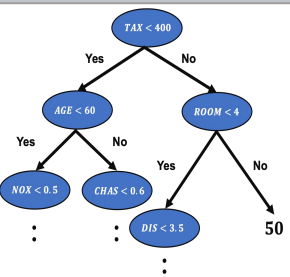
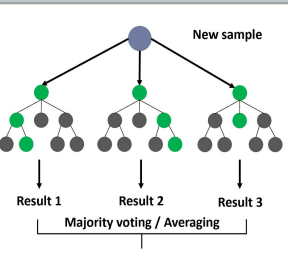
Laura, Alyssa, Wendi

Workflow

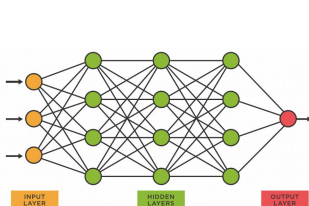
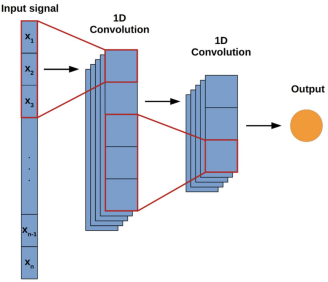
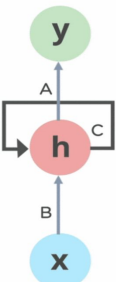
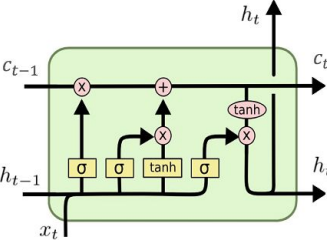
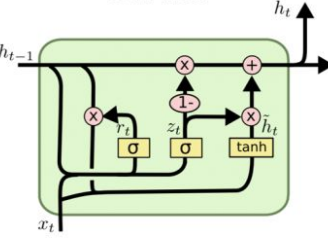
- Data
 - WoS 2021 spreadsheet
 - title + abstract → label
- Vectorization
 - preparing text for model
 - Bag-of-words, TFIDF, word vectors
- Model
 - Linear, NN, transformer
 - Output: relevant/not
- Evaluation:
 - Precision: true positives/relevant elements
 - Recall: false positives/retrieved elements
 - F1: harmonic mean (punishes lower scores more)



Linear Models

Support Vector Machine	K Nearest Neighbors	Decision Tree	Random Forest	Logistic Regression	Naive Bayes
				$\hat{p}(X_i) = \frac{1}{1 + \exp(-X_i w - w_0)}$	$P(y \mid x_1, \dots, x_n) = \frac{P(y)P(x_1, \dots, x_n \mid y)}{P(x_1, \dots, x_n)}$
Finds best decision boundary between classes	Decides class based on closest other points	Makes splits in data to best decide class	Ensemble of multiple decision trees	Calculates class probability by variables	Calculates class probability by variables, assuming independence
Prec: 0.24 Recall: 0.65 F1: 0.35	Prec: 0.39 Recall: 0.14 F1: 0.20	Prec: 0.25 Recall: 0.31 F1: 0.28	Prec: 0.09 Recall: 0.24 F1: 0.15	Prec: 0.27 Recall: 0.69 F1: 0.39	Prec: 0.25 Recall: 0.79 F1: 0.38

Neural Networks

Basic NN	CNN	RNN	LSTM	GRU
				
Linear algebra of nodes*weights put through a non-linear fcn after each layer	Sliding window of some function over nodes	Runs each input through the layers sequentially	Has “gates” to choose info to remember, forget, and use for current input, better memory than RNN	Combines remember and forget gates, more memory efficient than LSTM
	Prec: 0.24 Recall: 0.26 F1: 0.25	Prec: 0.13 Recall: 0.06 F1: 0.08	Prec: 0.14 Recall: 0.21 F1: 0.17	Prec: 0.17 Recall: 0.41 F1: 0.24

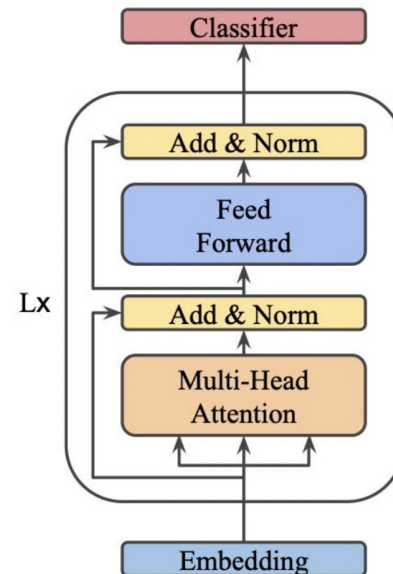
Transformers

BERT (Bidirectional Encoder Representations from Transformers)

- Contextualizes text by looking at both preceding and following words
- Trained off of English Wikipedia + Book Corpus
- DistilBERT, TinyBERT, SciBERT

SPECTER

- Designed to create document-level embeddings for scientific documents
- Uses the citation graph during pre-training → Incorporates information about inter-document relatedness
- Built off of SciBert & large corpus of scientific text

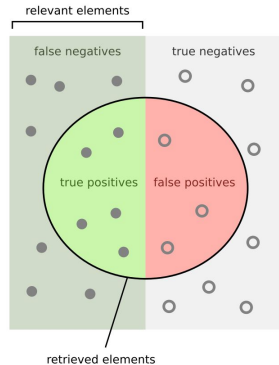


BERT Model	Prec.	Recall	F1
normal	0.21	0.86	0.34
mini	0.28	0.68	0.40

Summary & Next Steps

Best models so far

Model	Precision	Recall	F1
Naive Bayes (linear)	0.25	0.79	0.38
BERT (transformer)	0.20	0.86	0.34



How many retrieved items are relevant?

$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

How many relevant items are retrieved?

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

- Hyperparameter tuning
 - Word vectorization
 - Model size
 - Dropout, learning rate, etc
- More data available?
- Any notes from Susan?
 - Model preference? (simple/complex)
 - Metric to prioritize? (precision/recall)

Questions?